Mark as done

This lesson explains what are measures and how we can create them.

Lesson is currently being previewed. ×

Measures

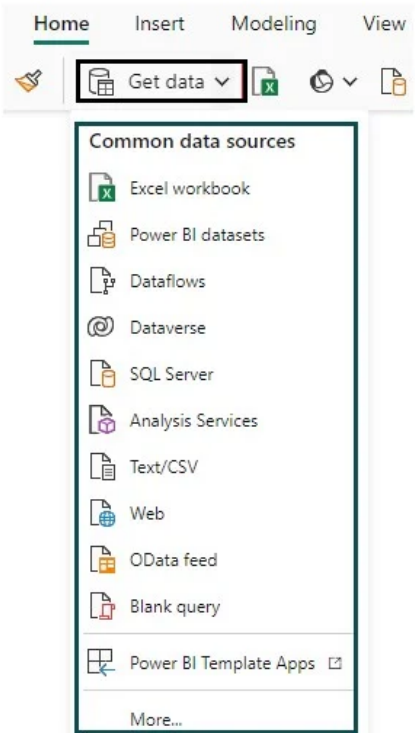
What are Measures in Power BI?

Measures in Power BI are calculations or aggregations you create to process your data, such as sum, average, count, or any custom calculation. Measures are often used in Power BI to derive insights from your data and create meaningful visualizations. They are written using Data Analysis Expressions (DAX), a formula language specific to Power BI and other Microsoft products like Excel.

How to Create Measures in Power BI?

To create new Power BI measures, follow the steps highlighted below:

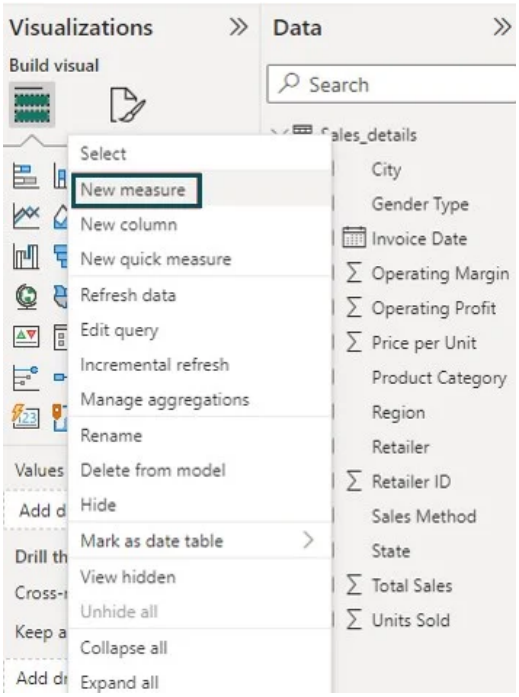
Step 1: Open your Power BI Desktop application. Load or import your data source into Power BI.



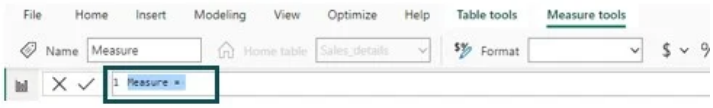
Step 2: Once imported into Power BI, navigate to the **Fields** pane to view the data attributes of the dataset.

Step 3: Select the table you want to create a measure for. Right-click on the table and choose the **New measure** option.

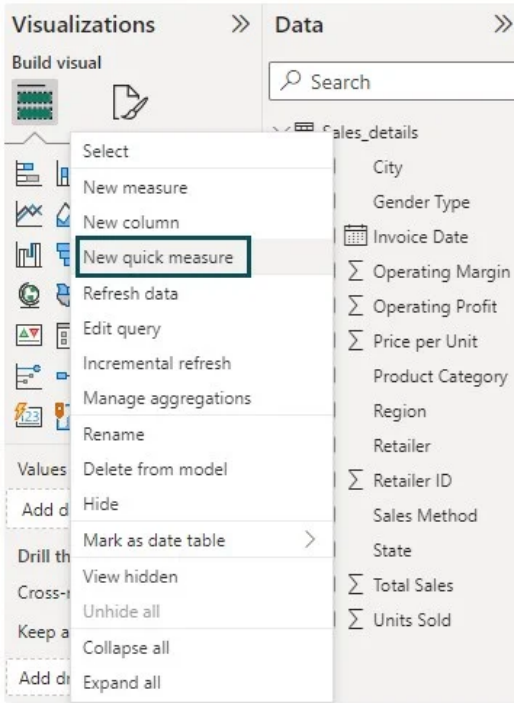
?



Step 4: Write your DAX measures in Power BI in the formula bar at the top. Give it a descriptive name and Press Enter to create the measure.

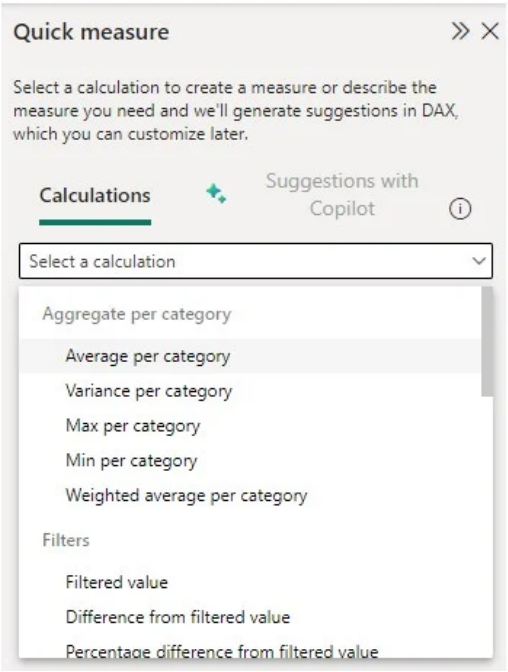


Step 5: You can also use the **New quick measure** option to create new measures. Select the **New quick measure** option when you right-click on the table you want to create a measure for.



It will open a **Quick measure** window where you can choose the predefined measures for your visualization requirements.

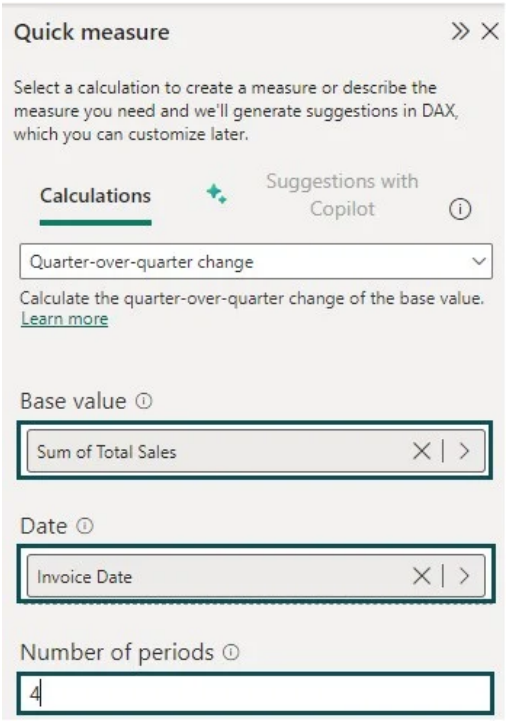
Step 6: Select a calculation to create a measure for your visual from the **Select a calculation** dropdown.



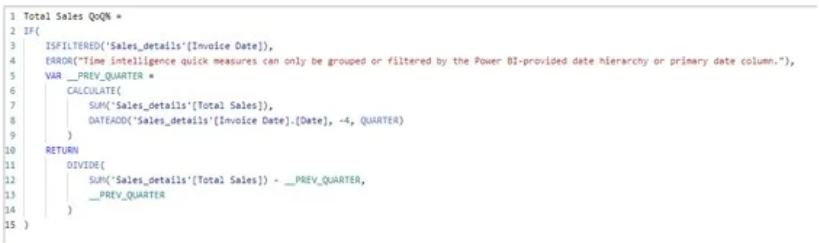
Power BI quick measure supports multiple in-built measures ranging from **Aggregate per category, Filters, Time intelligence functions, Totals, and Mathematical operations** functions to be used across the visuals.

Once you choose a measure, Power BI generates its DAX expression and creates a measure for your table.

Step 7: Choose a calculation and specify the sections for the measures, such as **Base value, Date, and Number of periods**. These sections may vary depending on your chosen calculation from the Calculations dropdown.

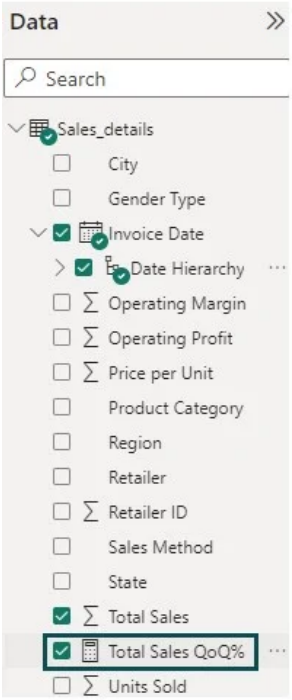


It will generate a DAX expression for the chosen measure in Power BI.



```
Total Sales QoQN =
IF(
    ISFILTERED('Sales_details'[Invoice Date]),
    ERROR("Time intelligence quick measures can only be grouped or filtered by the Power BI-provided date hierarchy or primary date column."),
    VAR __PREV_QUARTER =
        CALCULATE(
            SUM('Sales_details'[Total Sales]),
            DATEADD('Sales_details'[Invoice Date].[Date], -4, QUARTER)
        )
    RETURN
        DIVIDE(
            SUM('Sales_details'[Total Sales]) - __PREV_QUARTER,
            __PREV_QUARTER
        )
)
```

Step 8: Navigate to the **Fields** pane to view the created measure for the table



Note: You can also create dynamic measures in Power BI as per user interaction through slicers using functions like SWITCH, IF-ELSE, etc.

Similarly, you can also create group measures in Power BI. For this, you can create display folders and drag and drop your measures into the folder.

In the next section, we will see how to use new Power BI measures.

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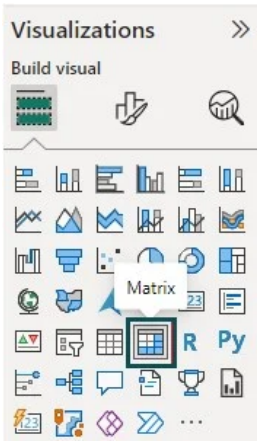
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How to Use Measures in Power BI?

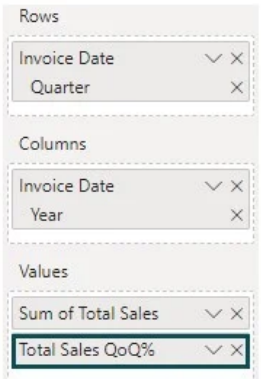
Once you've created measures, you can use them in visuals and calculations.

To use new measures in Power BI, follow the steps highlighted below:

Step 1: Navigate to the **Visualizations** pane and select a visual icon (For example Pie chart, Donut chart, stacked column chart, etc.)



Step 2: Navigate to the **Fields** pane and drag and drop the field attributes, including the measures created on the dataset table, to field sections of the visual or the report canvas.



Step 3: Customize your visuals by adding dimensions, filters, or slicers as needed.

Step 4: Refresh your report to see updated results when the data changes.

Examples

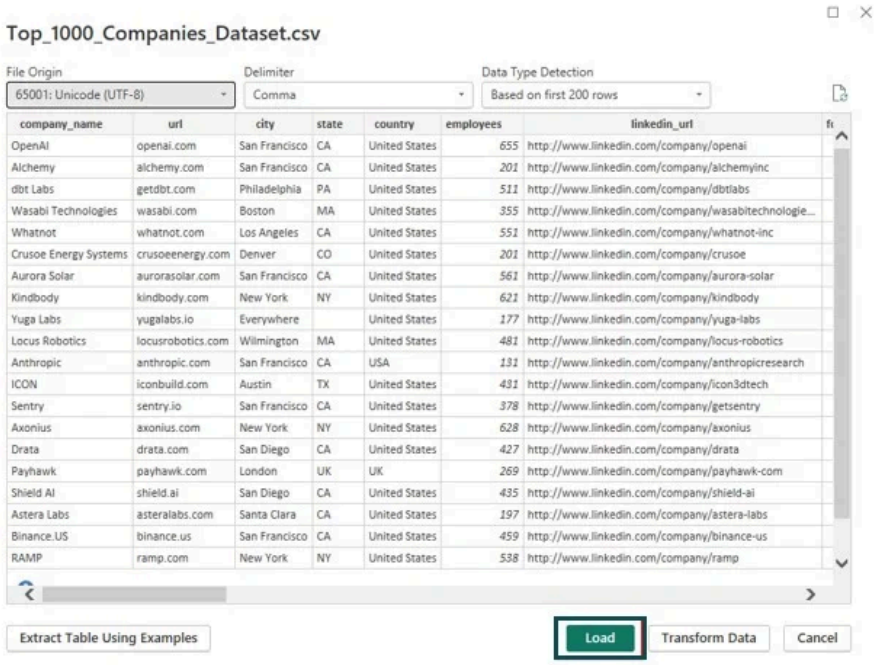
This section will demonstrate how to create and use new measures in Power BI in a step-by-step process with two examples.

Example #1

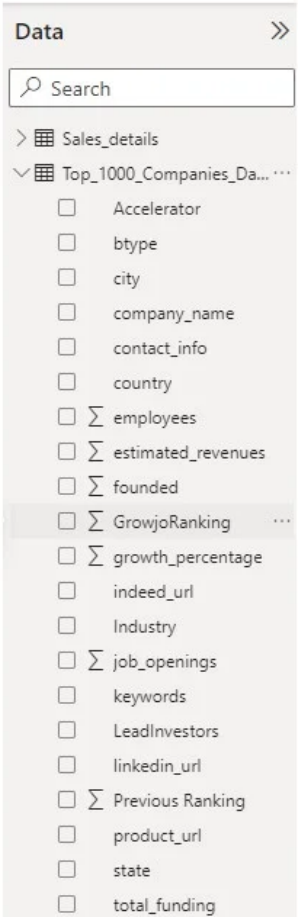
Here, we will create and use a measure in Power BI showing the total valuation of the Top 100 companies in the world using the **Top_1000_Companies_Dataset** data. This dataset contains the details of the Top 1000 global companies along with their funding and valuation information.

To create a measure in Power BI, follow the steps highlighted below:

Step 1: Open the Power BI Desktop, import the **Top_1000_Companies_Dataset** dataset using the **Get Data** option, and click on the **Load** button.

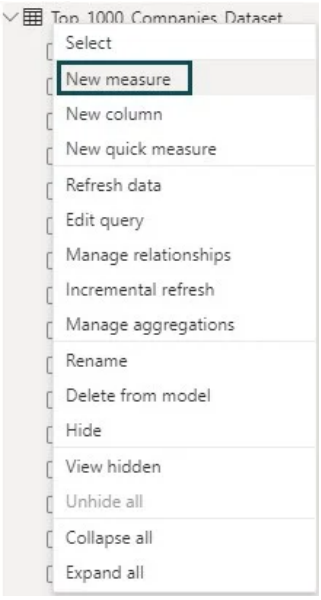


Once the data is loaded, you shall be able to view it by navigating to the **Fields** pane.



Step 2:

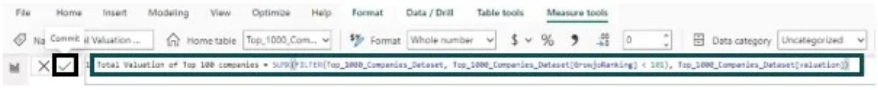
- Select the **Top_1000_Companies_Dataset** table.
- Right-click on it.
- Select the **New measure** option from the context menu.



Step 3: Write your DAX measure in Power BI in the formula bar at the top. Give your measure a descriptive name and Press Enter or Click on the **Commit** icon to create the measure.

Here, we have created the **Total Valuation of Top 100 companies** measure by filtering the Top 100 companies and summing their valuations.

Total Valuation of Top 100 companies = SUMX(FILTER(Top_1000_Companies_Dataset, Top_1000_Companies_Dataset[GrowjoRanking] < 101), Top_1000_Companies_Dataset[valuation])



You can view the measure in the **Fields** pane.



Step 4:

- Navigate to the **Visualizations** pane.
- Select the **Card** visual icon.
- Drag and drop the measure created to the **Fields** section.



It will create a Card visual showing the total valuation of the Top 100 companies in the world.

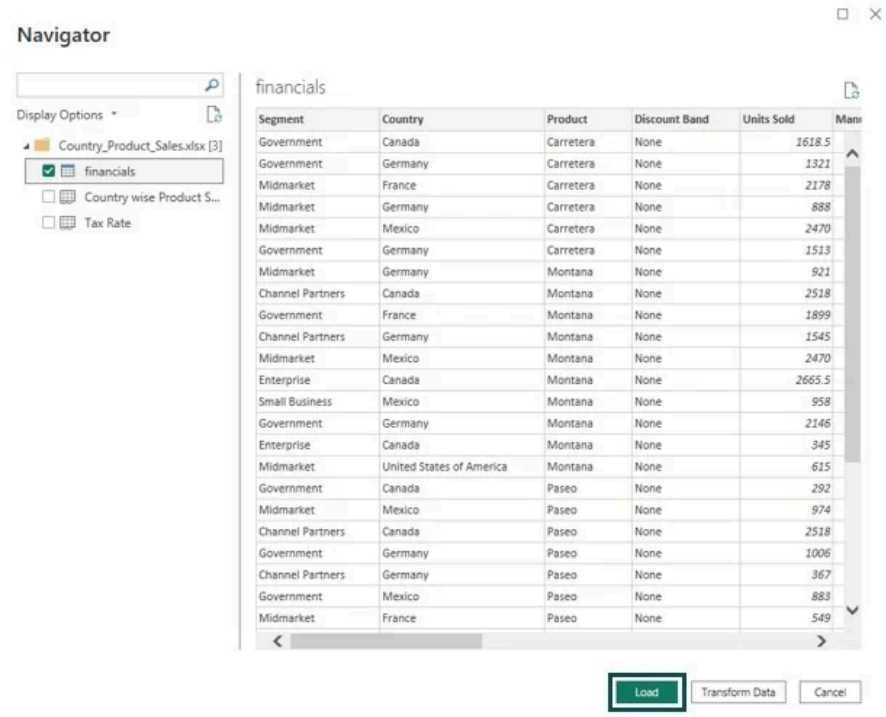


Example #2

In this example, we will create a measure of Sales YoY% and use it in data visualization using the **Country_Product_Sales** dataset. **Country_Product_Sales** dataset contains the details of sales data across different geographies, products, and segments.

To create and use measures, follow the steps highlighted below:

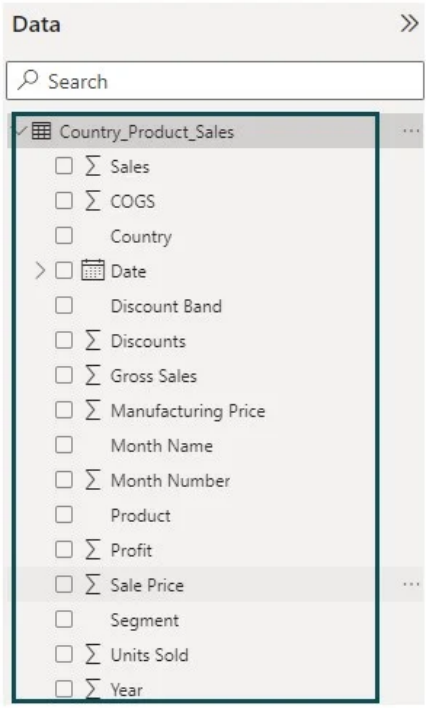
Step 1: Open the Power BI Desktop, import the **Country_Product_Sales** dataset using the **Get Data** option, and click the **Load** button.



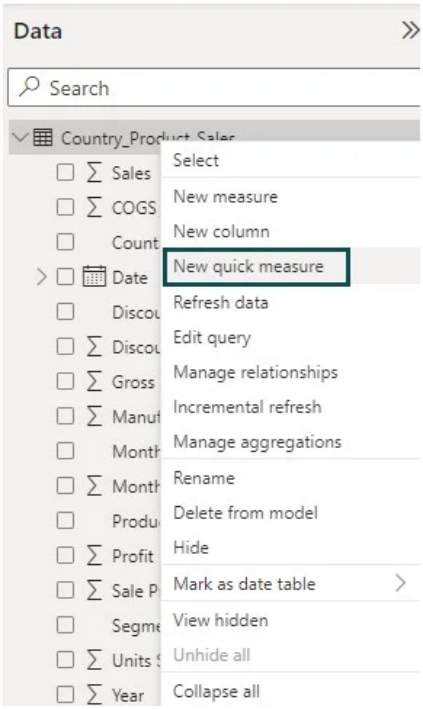
Segment	Country	Product	Discount Band	Units Sold	Manı
Government	Canada	Carretera	None	1618.5	
Government	Germany	Carretera	None	1321	
Midmarket	France	Carretera	None	2178	
Midmarket	Germany	Carretera	None	888	
Midmarket	Mexico	Carretera	None	2470	
Government	Germany	Carretera	None	1513	
Midmarket	Germany	Montana	None	921	
Channel Partners	Canada	Montana	None	2518	
Government	France	Montana	None	1899	
Channel Partners	Germany	Montana	None	1545	
Midmarket	Mexico	Montana	None	2470	
Enterprise	Canada	Montana	None	2665.5	
Small Business	Mexico	Montana	None	958	
Government	Germany	Montana	None	2146	
Enterprise	Canada	Montana	None	345	
Midmarket	United States of America	Montana	None	615	
Government	Canada	Paseo	None	292	
Midmarket	Mexico	Paseo	None	974	
Channel Partners	Canada	Paseo	None	2518	
Government	Germany	Paseo	None	1006	
Channel Partners	Germany	Paseo	None	367	
Government	Mexico	Paseo	None	883	
Midmarket	France	Paseo	None	549	

You can also use the **Transform Data** option to apply any data transformation or cleaning.

Once the data is loaded, you shall be able to view it by navigating to the **Fields** pane.



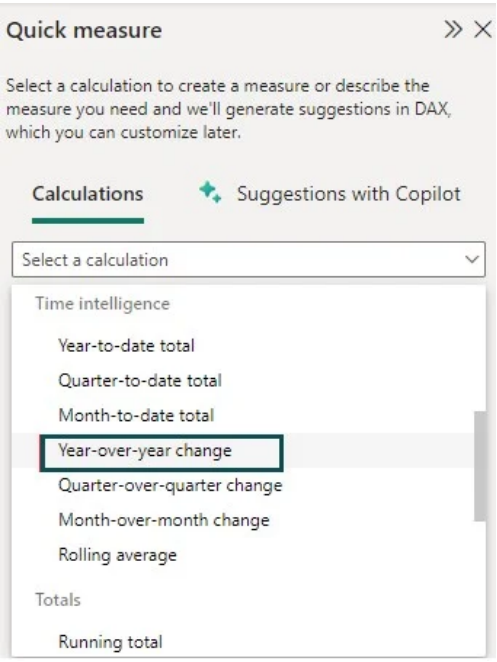
Step 2: Select the **Country_Product_Sales** table, right-click on the **Country_Product_Sales** table, and select the **New quick measure** option from the context menu.



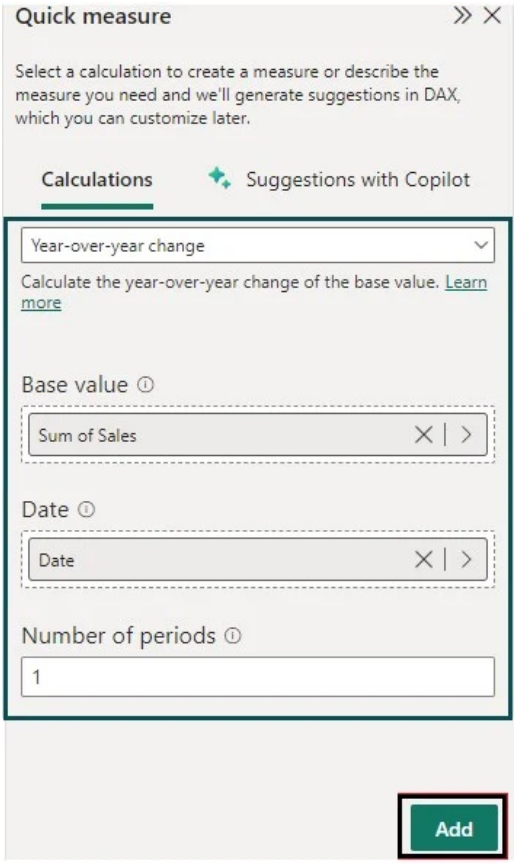
It will open a Quick measure window with built-in measures you can choose from.

Step 3: Select a calculation to create a measure for your visual from the **Select a calculation** dropdown on the **Quick measure** window.

Here we have chosen **Year-over-year change** to assess the overall Sales change across all the products and segments.



Step 4: Specify the measure details such as **Base value, Date, and Number of Periods** sections by dragging and dropping the fields from the Fields pane. Click on the **Add** button once you have provided the data fields.



Note: These sections may vary depending on the measure you choose from Select a calculation dropdown.

Step 5: Once the changes are added, a new measure will be created for the **Country_Product_Sales** dataset.

Automatic DAX measures in Power BI are created based on your chosen measure. You can see the DAX expression by selecting the measure in the **Fields** pane.

[Sales YoY%] =

IF(

ISFILTERED('Country_Product_Sales'[Date]),

ERROR("Time intelligence quick measures can only be grouped or filtered by the Power BI-provided date hierarchy or primary date column."),

VAR __PREV_YEAR =

CALCULATE(

SUM('Country_Product_Sales'[Sales]),

DATEADD('Country_Product_Sales'[Date].[Date], -1, YEAR)

)

RETURN

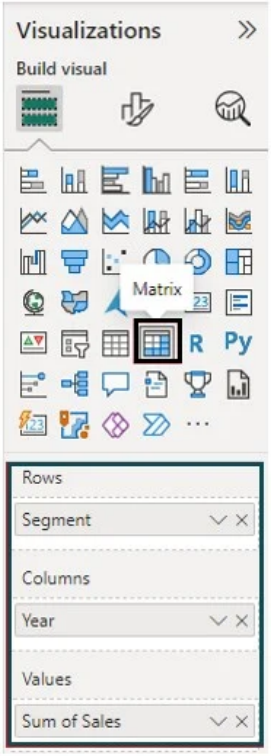
DIVIDE(SUM('Country_Product_Sales'[Sales]) – __PREV_YEAR, __PREV_YEAR)

)

```
1  [ Sales YoY%] =
2  IF(
3      ISFILTERED('Country_Product_Sales'[Date]),
4      ERROR("Time intelligence quick measures can only be grouped or filtered by the Power BI-provided date hierarchy or primary date
   column."),
5      VAR __PREV_YEAR =
6          CALCULATE(
7              SUM('Country_Product_Sales'[ Sales]),
8              DATEADD('Country_Product_Sales'[Date].[Date], -1, YEAR)
9          )
10     RETURN
11         DIVIDE(SUM('Country_Product_Sales'[ Sales]) - __PREV_YEAR, __PREV_YEAR)
12 )
```

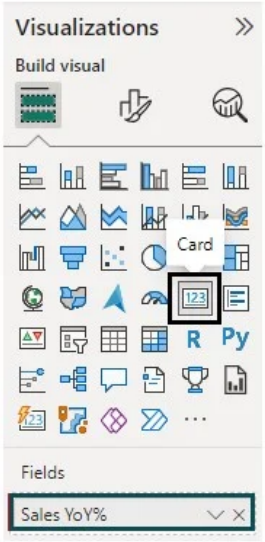
Step 6:

- Navigate to the **Visualizations** pane.
- Choose the **Matrix** visual icon.
- Drag and drop the fields from the **Fields** pane to the respective sections, such as **Rows, Columns, and Values**, as shown below.

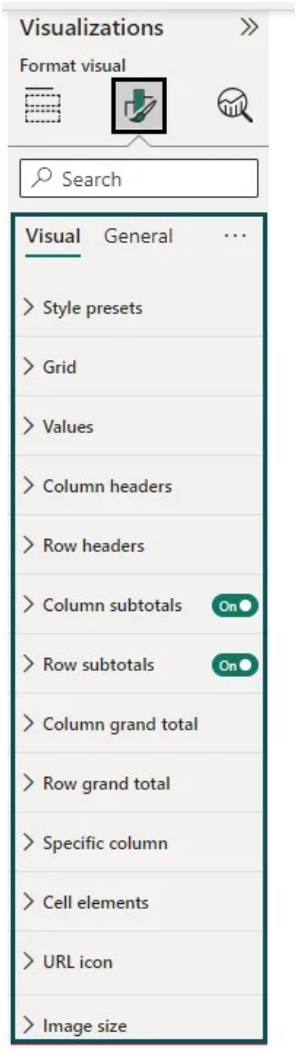


Step 7: Navigate to the **Visualizations** pane, choose the **Card** visual icon, and drag and drop the created measure from the **Fields** pane to the Fields section.

Here we have mapped the **Sales Yoy%** measure to the Fields section.



Step 8: Apply formatting to the visuals by navigating to the **Format your visual** tab under the **Visualizations** pane.



Post formatting, you can view the visual in the report canvas.

Segment	2013	2014	Total
Government	1,30,85,685.28	3,94,18,575.39	**5,25,04,260.67**
Small Business	83,35,674.00	3,40,92,244.50	**4,24,27,918.50**
Enterprise	40,49,562.50	1,55,62,131.88	**1,96,11,694.38**
Midmarket	5,46,243.45	18,35,639.63	**23,81,883.08**
Channel Partners	3,98,090.28	14,02,503.36	**18,00,593.64**
Total	**2,64,15,255.51**	**9,23,11,094.75**	**11,87,26,350.26**

349.46%

Sales YoY%

Important Things to Note

- When you interact with visuals in Power BI, measures are calculated on the fly, so they always reflect the most current data.
- DAX is a powerful formula language for creating measures and calculated columns. Take the time to learn DAX functions to create complex calculations.
- You can use measures to create Key Performance Indicators (KPIs), track trends, calculate ratios, and much more.

Frequently Asked Questions (FAQs)

1. How to add two measures in Power BI?

To add two measures together in Power BI, you can create a new measure that sums or combines the two existing measures. Here's how you can do it:

Step 1: Open your Power BI Desktop and load the dataset.

Step 2: Navigate to the **Fields** pane, right-click on the table, and click on **New measure** option to create measures.

Step 3: Enter the DAX expression in the formula bar to create measures and hit the Enter button. It will create measures on the selected table. You can follow these steps to create 2 measures.

Step 4: Navigate to the Fields pane, right-click on the table, and click on the **New measure** option.

Step 5: In the formula bar, enter the DAX expression to add those 2 measures you have created in the above steps
For example, to add Measure1 and Measure2, you can use the following formula:

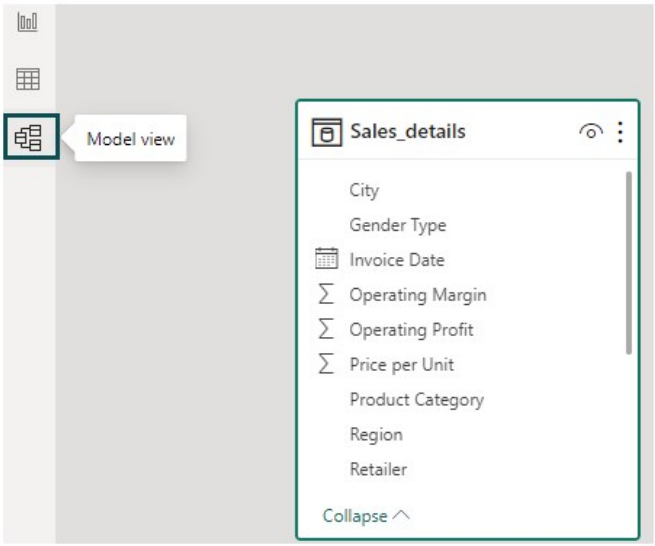
NewMeasure = [Measure1] + [Measure2]

Step 6: Press Enter to create the new measure in Power BI

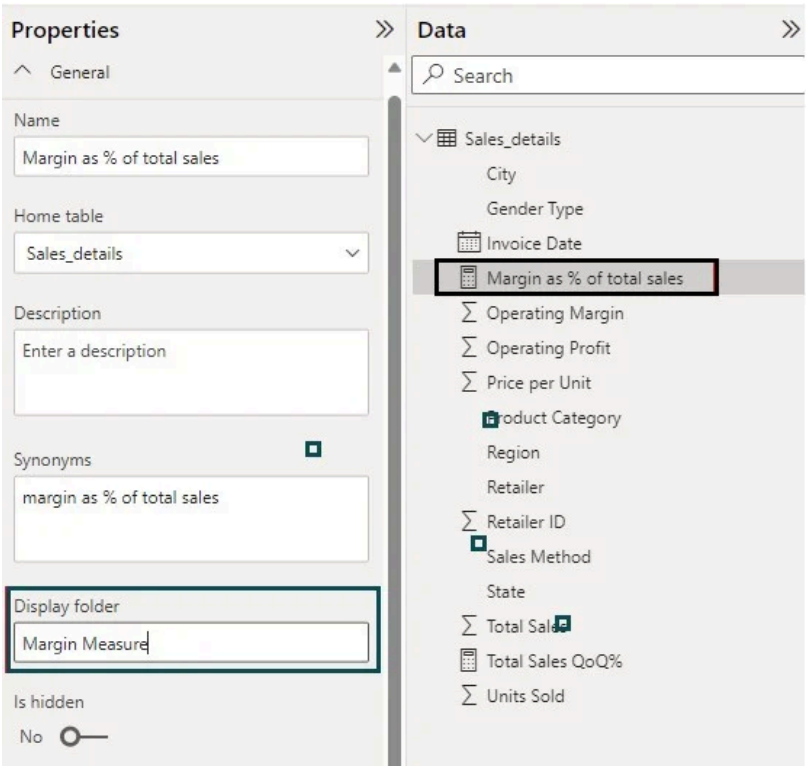
2. How to create a folder for measures in Power BI?

Organizing your measures into folders can help you manage your Power BI model more effectively. Here's how you can create a folder for measures:

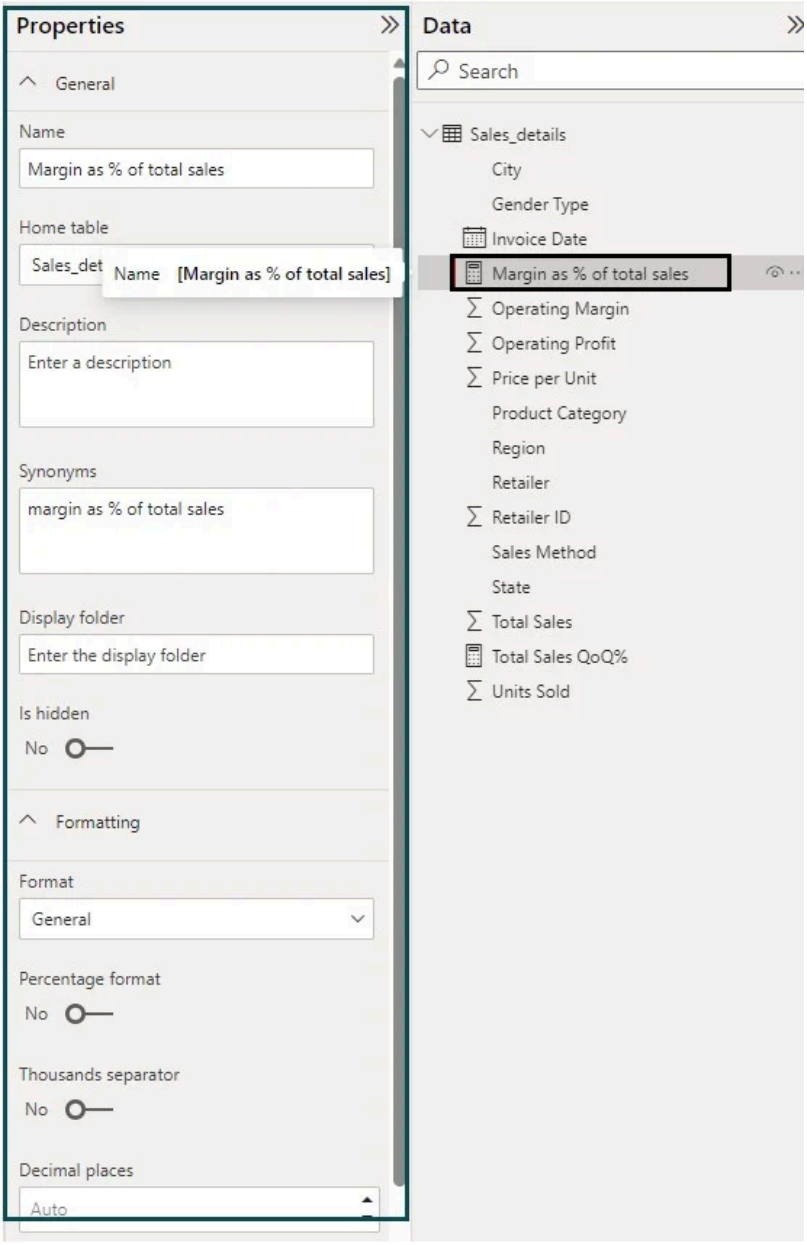
Step 1: Navigate to **Model** view from the left edge of the Power BI Desktop.



Step 2: Select the measure

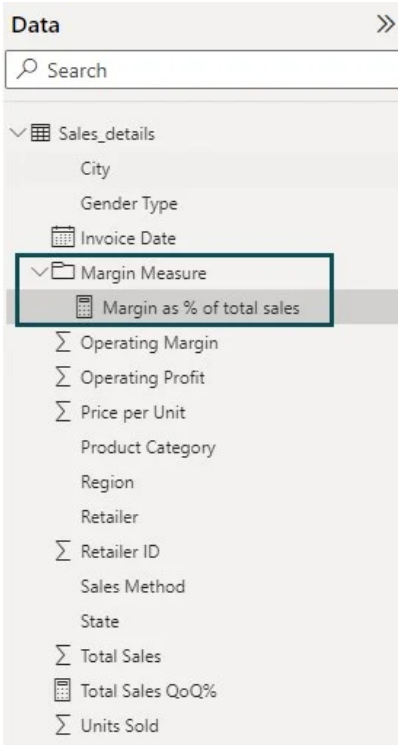


for which you want to create a folder from the **Fields** pane.



Step 3: In the **Properties** pane, enter a name for the new folder in the **Display folder** to create a folder.

Step 4: Navigate to the **Fields** pane to view the folder for the measure.



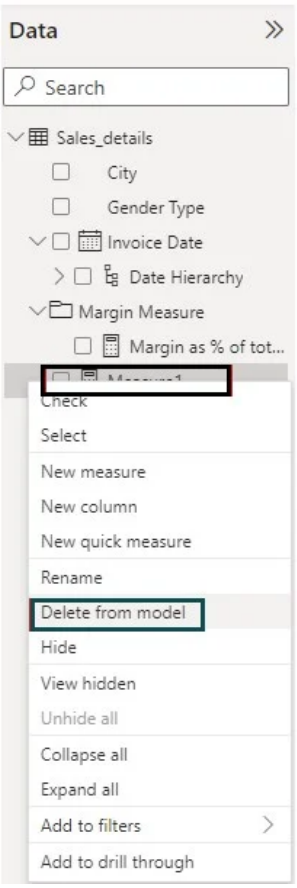
3. How to delete multiple measures in Power BI?

To delete multiple measures in Power BI, follow the steps highlighted below:

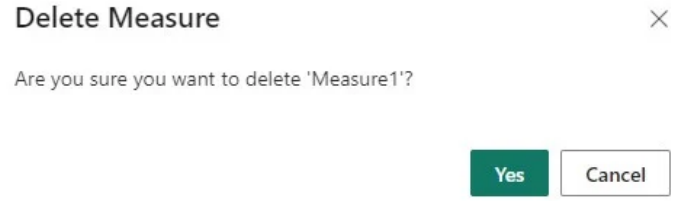
Step 1: Navigate to the **Fields** pane and select the table containing the measures you want to delete.

Step 2: Expand the table fields and select the measure you want to delete

Step 3: Right-click on the selected measure and choose **Delete from model** option from the context menu.



Step 4: Confirm the deletion in the prompt that appears by choosing Yes button.



Post confirmation, the measure will be deleted from the table. You can repeat these steps for multiple measures in Power BI.

4. How to sum measures in Power BI?

If you want to create a new measure that sums up existing measures, you can follow the steps mentioned in the first point for adding two measures. You can use simple addition to sum measures or you can use the SUMX function to sum multiple measures, depending on your specific requirements.

For example, if you have 2 measures **Total Sales Amazon** and **Total Sales Walmart** that you have created on the **Sales_details** table and you want to create a new measure that sums them, you can use the following DAX formula:

AddMeasure = SUMX(Sales_details,Sales_details[Total Sales Amazon] + Sales_details[Total Sales Walmart])

```
1 AddMeasure = SUMX(Sales_details,Sales_details[Total Sales Amazon] + Sales_details[Total Sales Walmart])
```

End of Lesson

You will not see the progress bar because you can edit this lesson